
Overview

This assignment has two components:

1. **Blackboard:**
You will select your own dataset and answer conceptual questions about preprocessing and prediction.
2. **GitHub Classroom:**
You will implement *three* core preprocessing functions and run a complete preprocessing pipeline using starting code provided to you.

Only the Blackboard component will be graded; you are *strongly encouraged* to attempt the GitHub Classroom component as future assignments will be formatted this way (think of it as a test drive/means for you to practice GitHub Actions).

PART 1 — Blackboard

Part 1 evaluates your understanding of why we preprocess data. You will choose your own dataset and answer conceptual questions.

Dataset Requirements (Important)

Choose a dataset that requires at least 3–5 preprocessing steps. Examples of common preprocessing needs:

- Missing values
- Categorical variables that require encoding
- Numeric features needing scaling
- Outliers
- Duplicate rows
- Columns that should be excluded from modeling, such as:
 - ID columns
 - Leakage variables (contain information unavailable at prediction time)

Note: Avoid “toy” datasets that do not require real preprocessing (e.g., Iris). Omit any columns that may contain sensitive information for privacy purposes (dataset depending).

Blackboard Questions

Multiple Choice / Matching (3 points total)

Three questions worth 1 point each assessing conceptual understanding.

Short Answer (12 points total)

Three questions total; more details in Blackboard, but briefly:

1. Upload a screenshot of your data. The screenshot must show column names and several rows.
2. Explain what you're trying to predict, and why. Include your target variable and explain why the prediction matters.
3. Explain what preprocessing steps your dataset requires.

Bullet points are encouraged! Address the following as they apply to your selected dataset:

- Which preprocessing steps apply
- Why they are needed
- Any target transformations
- Categorical encoding
- Scaling choices
- Missing data
- Outliers
- Column exclusions (IDs, leakage, etc.)

Note: This is conceptual only — you do not implement preprocessing here.

PART 2 — GitHub Classroom

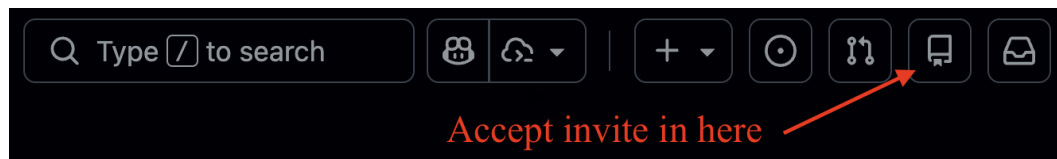
Goal

To help you gain confidence with core preprocessing tasks by implementing only the essentials, while the full workflow (splitting, saving, summaries, etc.) is handled for you.

What You Will Do

Click on the assignment [link](#) (also linked in Blackboard). As this is your first time interacting with the Classroom interface, you will need to link your GitHub account. Find your name in the list and follow any sign-in instructions. Once your account is linked, a repository for the assignment will be created for you.

IMPORTANT: you need to accept the invitation to the assignment before you can view/edit it. For this, you can navigate to GitHub. The notification to accept the invite is in your 'Notifications' tab.



Once the permissions are sorted, you will be able to view your GitHub Classroom repository that includes:

Functions That You Will Code

1. `detect_feature_types()`
Identify categorical vs numeric columns.
2. `encode_categorical()`
Apply one-hot encoding to categorical features, track new columns.
3. `scale_numeric()`
Apply standardization (z-score scaling) to numeric columns.

What's Already Provided

- ✓ `replace_missing_values()`
- ✓ `remove_duplicates()`
- ✓ A complete main preprocessing pipeline (`run_preprocessing()`) that handles:
 - Data loading and validation
 - Missing value handling
 - Train/test splitting (with stratification)
 - Calling your three functions
 - Ensuring train/test column consistency
 - Saving processed files
 - Producing summary statistics

Note: You do not implement file I/O, splitting, orchestration, or advanced error handling; this has been intentionally simplified so you can focus on the preprocessing fundamentals.

Testing & Autograding

Tests run automatically when you push your code to your repository.

AI Usage Disclosure (MANDATORY)

Your repository must contain a completed `AI_DISCLOSURE.md` file.

You must disclose:

- Any AI tools used
- What tasks they assisted with
- How you verified correctness

Note: If you do not complete the AI usage disclosure honestly and fully, you may receive zero marks for Part 2, regardless of test performance. Relevant for future assignments only.

Submission

You do not upload anything to Blackboard for Part 2. Your submission is complete when:

- Your functions are implemented
 - All tests have run on GitHub
 - The AI Usage Disclosure is complete
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GRADING SUMMARY

Part 2 — Blackboard (100%)

- MC/matching (3 points)
- Short answer (12 points)

Total Blackboard = 15 points → 100%

Part 2 — GitHub Classroom (not graded)

- Correctly implemented 3 required functions
 - Passing autograding tests
 - Demonstration in the notebook
 - Completed AI Usage Disclosure (required)
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Integrity, Expectations, & Advice

Use of AI Tools

AI can help you brainstorm or check understanding, but it cannot replace your own reasoning. Completing the AI Usage Disclosure is mandatory.

Tips for Success

- Start with TUTORIAL.ipynb in the GitHub repo
- Review provided examples carefully
- Implement one function at a time
- Push early and often to see test feedback
- Don't overthink Part 1 → it's about understanding, not coding